MIL-T-10086H
8 August 1980
SUPERSEDING
MIL-T-10086G
5 May 1975

MILITARY SPECIFICATION

TANKS, LIQUID STORAGE, METAL;

VERTICAL, BOLTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers vertical, cylindrical, bolted steel tanks for on-the-ground storage of petroleum products; on-the-ground or tower-supported water storage; fitting sets to convert the 100, 250, 500, 1,000, and 3,000 barrel (bbl) tanks from on-the-ground storage to tower-supported water storage; and a 100-bbl tank fitted for on-the-ground asphalt storage.

1.2 Classification. Tanks covered by this specification shall be the following types and nominal capacities (42 gal/bbl), as specified (see 6.2):

Type I - Petroleum products and potable storage.
Type II - Petroleum products and nonpotable water storage.

100 bbl (4,200 gal)	1,000 bbl (42,000 gal)
250 bbl (10,500 gal)	3,000 bbl (126,000 gal)
500 bbl (21,000 gal)	10,000 bbl (420,000 gal)

Type III - Asphalt storage 100 bbl (4,200 gal).

FSC 5430

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-17T13:32Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in fect on date of invitation for bids or request for proposal form a rt of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

H-B-420	- Brush, Paint, Flat, Metal-Bound.
QQ-S-781	- Strapping, Steel, and Seals.
TT-C-494	- Coating Compound, Bituminous, Solvent type, Acid Resistant.
TT-V-121	- Varnish, Spar, Water-Resisting.
UU-T-81	- Tags, Shipping and Stock.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.
PPP-T-60	- Tape: Packaging, Waterproof.

MILITARY

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-B-131	- Barrier Materials, Watervaporproof, Flexible, Heat-Sealable.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-1086	- Gasket Material, Synthetic Rubber (for Bolted Steel Tanks).
MIL-L-2638	- Lacquer, Vinyl Resin, Gasoline and Water Resistant.
MIL-C-4556	- Coating Kit, Epoxy, For Interior of Steel Fuel Tanks.
MIL-P-13983	- Paint, Temporary, Lusterless, Gasoline Removable.
MIL-S-14231	- Sealing Compound, Joint, Two Component, for Bolted Aluminum or Steel Petroleum Storage Tanks.
MIL-P-20628	- Putty, Sealing (for Bolted Steel Storage Tanks).

Source: https://assist.dla.mil -- Downloaded: 2026l0-17T13:32Z
Check the source to verify that this is the currentversion before use.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

DEPARTMENTS OF THE ARMY AND THE AIR FORCE MANUAL

TM5-5430-209-12 TO 36Y31-1-101	- Tank, Steel, Vertical, Bolted, Knockdown, Sealed Openings, Standard Bottom and Roof, Gasoline, Oil, or Water.

DRAWINGS

ME

TA13216E4940	- Tank, Asphalt Storage Steel, Vertical Bolted, 100 bbl.
TA13216E7200	- Tank, Liquid Storage, Steel, Vertical Bolted, 100 bbl.
TA13216E7300	- Tank, Liquid Storage, Steel, Vertical Bolted, 250 bbl.
TA13211E8025	- Tank, Liquid Storage, Steel, Vertical Bolted, 500 bbl.
TA13211E8050	- Tank, Liquid Storage, Steel, Vertical Bolted, 1,000 bbl.
TA13216E7942	- Tank, Liquid Storage, Steel, Vertical Bolted, 3,000 bbl.
TA13211E8100	- Tank, Liquid Storage, Steel, Vertical Bolted, 10,000 bbl.
TA13216E8260	- Fitting Set, Asphalt Tank 100 bbl Assembly.
TA13216E8261	- Fitting Set, Water Tower, 100 bbl.
TA13216E8262	- Fitting Set, Water Tower, 250 bbl.
TA13216E8263	- Fitting Set, Water Tower, 500 bbl.
TA13218E5028	- Fitting Set, Water Tower 1,000 bbl.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

TA13218E5029	- Fitting Set, Water Tower, 3,000 bbl.
D13217E2317-1 through -6	- Packing for Overseas Shipment - Tank, Steel, Vertical, Bolted, Knockdown, Convertible, 100 bbl.
D13217E2318-1 through -6	- Packing for Overseas Shipment - Tank, Asphalt Storage Steel, Vertical Bolted, 100 bbl.
D13217E2319-1 through -5	- Packing for Overseas Shipment - Tank, Steel, Vertical, Bolted, Knockdown, Convertible, 250 bbl.
D13217E2320-1 through -7	- Packing for Overseas Shipment - Tank, Steel, Vertical, Bolted, Knockdown, Convertible, 500 bbl.
D13217E2321-1 through -10	- Packing for Overseas Shipment - Tank, Steel, Vertical, Bolted, Knockdown, Convertible, 1,000 bbl.
D13217E2322-1 through -14	- Packing for Overseas Shipment - Tank, Steel, Vertical, Bolted, Knockdown, Convertible, 3,000 bbl.
D13217E2323-1 through -20	- Packing for Overseas Shipment - Tank, Steel, Vertical, Bolted, Knockdown, 10,000 bbl.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN WELDING SOCIETY (AWS)

D1.1 Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. Seventh Street, Miami, FL 33125.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Code for Unfired Pressure Vessels, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

Source: https://assist.dla.mil -- Downloaded: 20260-17T13:32Z
Check the source to verify that this is the current version before use.

STEEL STRUCTURES PAINTING COUNCIL (SSPC)

SSPC-SP-6-63 - Surface Preparation Specification No. 6, Commercial Blast Cleaning.

(Application for copies should be addressed to the Steel Structures Painting Council, 4400 Fifth Avenue, Pittsburgh, PA 15213.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The tanks and fitting sets shall be in accordance with applicable top assembly drawings and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the tank. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (first produced tank). The contractor shall furnish one or more tanks as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove that his production methods will produce tanks that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4, and shall be subject to surveillance and approval by the Government (see 6.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

3.3 Initial production inspection. When specified (see 6.2), the contractor shall furnish to the Government one or more tanks for inspection as specified in 4.4.

3.4 Materials. Materials shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.14). When the tanks are intended for the storage of potable water, all of the materials with which potable water will come in contact shall have no adverse effect on personnel. Questions pertinent to this effect shall be referred to the appropriate Department medical service officer who will act as advisor to the contracting officer (see 6.8).

3.5 Sealing gaskets. Unless otherwise specified herein, sealing gaskets shall be as shown on Drawing B13211E8056 and shall conform to MIL-G-1086, Type II. When specified (see 6.2), stepped type gaskets shall be furnished and these gaskets shall conform to Figure 1. The stepped type gaskets shall be molded or extruded from polychloroprene synthetic rubber conforming to MIL-G-1086, Type II, except that the low temperature compression set at minus 20° F, shall be not more than 75 percent.

3.6 Sealing compound, putty, and brushes. When specified (see 6.2), sealing compound, putty, and brushes shall be furnished in the following quantities for the tank capacity specified (see 6.5):

Tank Capacity, Barrels	100	250	500	1M	3M	10M
Sealing Compound, quart	3	6	15	28	28	68
Putty, gal.	1	1	1	2	2	4
Brush, each	4	4	4	6	6	6

3.6.1 Sealing compound and putty. All containers of sealing compound and putty shall be marked as follows:

"DO NOT USE IN CONTACT WITH POTABLE WATER"

3.6.2 Sealing compound. Sealing compound shall conform to MIL-S-14231.

3.6.3 Putty. Putty shall conform to MIL-P-20628.

3.6.4 Brushes. Brushes shall conform to H-B-420, Type 1, Grade C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

3.7 Gaging devices. The automatic liquid-level indicating gage shown on Drawing D13200E9129 shall be furnished with installation and operating instructions, except that automatic liquid-level indicating gage shall not be supplied with Type III, asphalt storage tanks.

3.8 Identification marking. The tanks shall be identified in accordance with MIL-STD-130, and as specified herein.

3.9 Cleaning, treatment and painting.

3.9.1 Cleaning. All ferrous metal surfaces requiring painting shall be cleaned to the bare metal by blasting in accordance with Steel Structures Painting Council Specification SSPC-SP-6-63.

3.9.2 Treatment and painting.

3.9.2.1 Exterior. The exterior of the tanks, except the underlapping portion of the chine over chine joints; and all fittings, except those installed inside the tanks; shall be treated and painted in accordance with MIL-T-704, Type F.

3.9.2.2 Interior.

3.9.2.2.1 Treatment. The interior of the tanks including the underlapping portion of the chine over chine joints and exposed surfaces of fittings installed inside the tank shall be treated and prepared for painting in accordance with MIL-T-704.

3.9.2.2.2 Painting. Unless otherwise specified herein (see 3.13.3), the interiors of the tanks shall be coated with a three-coat system of vinyl resin lacquer conforming to MIL-L-2638, Type I (gray) first coat, Type II (white) second coat, and Type III (gray) third coat, applied to a minimum dry film thickness of 1, 2, and 1 mils respectively to make a total dry film thickness of 4 mils.

3.10 Stenciling. All unassembled components of tanks and fitting sets shall be stenciled with erection marking or with applicable special marking as specified herein. Large components shall be stenciled with lusterless white enamel overcoated with varnish conforming to TT-V-121 or stenciled with paint conforming to MIL-P-13983. The letters shall be Gothic capitals and numerals and shall be 2 inches high where space permits. On small components, the stenciling shall be as large as practicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

3.10.1 Erection marking. Erection marking shall consist of the applicable drawing number. Small components shall have erection marking information on cloth tags conforming to UU-T-81, Type A or on corrosion-resistant metal tags. Cloth tags shall be waterproofed in accordance with MIL-STD-129. The tags shall be attached to the parts in such a manner as to prevent damage to the preservation and parts.

3.11 Erection manuals. Such technical publications as are specified shall be furnished (see 6.2). Erection manuals shall be as detailed in coverage as TM5-5430-209-12 for Army procurements and TO 36Y31-1-101 for Air Force procurements as applicable; copies of which will be furnished by the Government for use as a guide (see 6.6). The 100 bbl, 250 bbl, and 500 bbl capacity tanks shall each be furnished with one manual; the 1000 bbl and 3000 bbl capacity tanks each with two manuals; and 10,000 bbl capacity tanks each with three manuals.

3.12 Interchangeability. All like components of tanks of the same capacity shall be interchangeable with respect to assembly and fit. The contractor shall provide and use in-process gages necessary to control conformance to the drawings.

3.13 Type I and Type II tanks. The Type I and Type II tanks shall be in accordance with the following top assemblies for the nominal capacities specified:

Capacity	Top Assemblies
100 bbl (4,200 gal)	TA13216E7200
250 bbl (10,500 gal)	TA13216E7300
500 bbl (21,000 gal)	TA13211E8025
1,000 bbl (42,000 gal)	TA13211E8050
3,000 bbl (126,000 gal)	TA13216E7942
10,000 bbl (420,000 gal)	TA13211E8100

3.13.1 Tower fitting sets. When specified (see 6.2), a tower-fitting set shall be furnished which will convert the tank from on-the-ground installation to tower-supported water service. Fitting sets shall be in accordance with the following top assemblies for the nominal capacities specified. A tower-fitting set is not used with the 10,000 bbl capacity tank.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

Capacity	Top Assemblies
100 bbl	TA13216E8261
250 bbl	TA13216E8262
500 bbl	TA13216E8263
1,000 bbl	TA13218E5028
3,000 bbl	TA13218E5029

3.13.2 Type I tank. All materials for the Type I tank which will come in contact with potable water shall have no adverse effect on the health of personnel when the tank is used for the storage of potable water (see 3.9.2.2.1, 3.9.2.2.2, and 6.8).

3.13.3 Type II tank. When specified (see 6.2), the interior of the Type II tank shall be treated and prepared for painting as specified in 3.9.2.2 and coated with a two-coat system of epoxy type paint conforming to MIL-C-4556 to a dry minimum dry film thickness of 6 mils.

3.13.3.1 Marking. The tank shall be stenciled "DO NOT USE FOR POTABLE WATER". Letters shall be 2 inches high and shall be stenciled on the exterior of all deck and bottom plates and staves.

3.14 Type III tank. The Type III tank shall be in accordance with Top Assembly TA13216E4940 and shall be furnished with a fitting set in accordance with Top Assembly TA13216E8260. When specified (see 6.2), interior of Type III tank shall be coated with a coating conforming to TT-C-494, Type I.

3.14.1 Marking. The tank shall be stenciled "DO NOT USE FOR POTABLE WATER". Letters shall be 2 inches high and shall be stenciled on the exterior of all deck and bottom plates and staves.

3.15 500 bbl, 1000 bbl, 3000 bbl, and 10,000 bbl capacity tanks. Unless otherwise specified (see 6.2), these tanks shall be inspected for all characteristics shown on the drawings under QAP's, Major, AQL 1.0 percent defective using inspection equipment conforming to the Government inspection equipment design drawings. The inspection equipment will be loaned by the Government (see 6.7) or shall be fabricated and furnished by the contractor as specified (see 6.2).

3.15.1 Government-loaned property. When specified (see 3.15) the following property in the quantities indicated will be loaned by the Government (see 6.7):

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

ITEM	QUANTITY	DESCRIPTION	IDENTIFICATION
1	1 set	Inspection Equipment for Tank, Liquid Storage, Metal; Bolted, 500 bbl Capacity.	Quality Assurance Pamphlet IE-FB-P715-131
2	1 set	Inspection Equipment for Tank, Liquid Storage, Metal; Bolted, 1000/3000 bbl Capacity	Quality Assurance Pamphlet IE-FB-P715-132
3	1 set	Inspection Equipment for Tank, Liquid Storage, Metal; Bolted, 10,000 bbl Capacity.	Quality Assurance Pamphlet IE-FB-P715-133

3.15.2 Inspection equipment fabricated and furnished by the contractor. When specified (see 3.15), the contractor shall fabricate the inspection equipment in accordance with the Government inspection equipment design drawings included in the top assembly (see 6.11). The contractor shall furnish the contracting officer, prior to inspection of the first produced tank, with written certification from an independent laboratory that the inspection equipment has been calibrated and complies in every detail with the drawings.

3.16 Workmanship. All parts, components and assemblies of the tanks including castings, molded parts, bearings, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. External surfaces shall be free from burrs, sharp edges, and corners except when sharp edges and corners are required.

3.16.1 Steel fabrication. Steel used in the fabrication of the tanks shall provide original quality surface finish and shall be free from kinks and sharp bends. Steel having an eroded surface is not acceptable. The forming of material shall be done by methods that will not cause damage to the metal. Shearing and clipping shall be done neatly and accurately. Corners shall be square and true. Burned surfaces of flame-cut material shall be free of slag. Precautions shall be taken to avoid overheating; and heated metals should be allowed to cool slowly except where heat treatment is required. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape.

3.16.2 Welders and welding.

3.16.2.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

3.16.2.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subject to proof and service loading.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility of inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced drawings, specifications, and standards, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the first produced tank shall be cause for disassembly, in the presence of a Government representative, of the first produced tank to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first produced tank. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring inspection, the sampling plan, and the basis for acceptance and rejection (see 6.9 and 6.10).

Source: https://assist.dla.mil -- Downloaded: 202610-17T13:32Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First produced tank inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 First produced tank inspection.

4.3.1 Examination. The first produced tank shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection, or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests.

4.3.2.1 100 bbl and 250 bbl capacity tanks. The first produced tank shall be tested as specified in 4.6.2.1. The Type III first produced tank shall also be tested as specified in 4.6.2.2. Failure to pass any test shall be cause for rejection, or for performing the disassembly inspection specified in 4.1.2.

4.3.2.2 500 bbl, 1000 bbl, 3000 bbl, and 10,000 bbl capacity tanks. Each characteristic on the tank designated under QAP's as Major, AQL 1.0 percent defective shall be inspected with the inspection equipment specified in 3.15 and any discrepancy shall be cause for rejection. When inspection equipment is not required (see 3.15) the tank shall be tested as specified in 4.6.2.1 and failure to pass the test shall be cause for rejection, or for performing the disassembly inspection specified in 4.1.2.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production tanks will be selected at random by the Government from the tanks being produced by production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production tank shall not exclude the remaining tanks from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production tank to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production tank(s) and shall be cause for refusal by the Government to continue acceptance of production tanks until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on tanks previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all tanks accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination and test shall be as follows:

Lot size	Sample size
1 to 5	0
6 to 24	1
25 to 50	2

Samples shall be selected at random.

4.5.1.1 Unit of product. The unit of product shall be one tank.

4.5.1.2 Lot size. A lot shall consist of not more than 50 and not less than six tanks of the same type and capacity. When the total quantity of tanks being purchased is five or less, no quality conformance sample inspection is required.

4.5.2 Examination.

4.5.2.1 Individual. Each tank shall be examined for Defect 101 as specified in 4.6.1. Presence of one or more defects shall be cause for rejection of the tank.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

4.5.2.2 Samples. Samples selected in accordance with 4.5.1 shall be examined as specified in 4.6.1, except for Defect 101. The presence of one or more defects shall be cause for rejection of the lot.

4.5.3 Tests.

4.5.3.1 Individual for Type III tank only. Each Type III tank shall be tested as specified in 4.6.2.2. Failure to pass the test shall be cause for rejection of the tank.

4.5.3.2 Samples of 100 bbl and 250 bbl capacity tanks. Samples selected in accordance with 4.5.1 shall be tested as specified in 4.6.2.1. Failure of any sample to pass the test shall be cause for rejection of the lot.

4.5.3.3 Samples of 500 bbl, 1000 bbl, 3000 bbl, and 10,000 bbl capacity tanks. Each characteristic designated under QAP's as Major, AQL 1.0 percent defective on each sample tank selected in accordance with 4.5.1 shall be inspected with the inspection equipment specified in 3.15 and any discrepancy shall be cause for rejection of the lot. When inspection equipment is not required (see 3.15) each sample tank shall be tested as specified in 4.6.2.1 and failure of any sample to pass the test shall be cause for rejection of the lot.

4.6 Inspection procedure.

4.6.1 Examination. Unless otherwise specified (see 3.15) Government-loaned inspection equipment (see 3.15.1) shall be used in accordance with Government Quality Assurance Pamphlet included in the applicable TA's to inspect dimensions indicated on the applicable drawings for the 500 bbl, 1000 bbl, 3000 bbl, and 10,000 bbl capacity tanks.

101. Any part or component not in accordance with the QAP requirements as shown on the drawings.
102. Material not as specified.
103. Components missing or not as specified.
104. Sealing compound, putty, and brushes not as specified or not furnished in the quantities specified.
105. Automatic gage and gage board not as specified.
106. Any drawing requirement, other than those identified by QAP data, not as specified.
107. Installation and operating instructions for the automatic gage not provided.
108. Identification marking and erection markings illegible or missing.
109. Stenciling incorrect or missing.
110. Erection manuals not as specified.
111. Cleaning, treatment and painting not as specified.
112. Workmanship not as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

4.6.2 Tests.

4.6.2.1 Erection test. Erect the tank. The tank shall be erected using gaskets only. Sealing compound and putty shall not be used during this test. Erection of the tank shall be accomplished in strict accordance with procedures outlined in the erection manual. Inability to erect the tank in accordance with erection manual procedures or nonmating of components shall constitute failure of this test.

4.6.2.2 Hydrostatic test. Subject the steam coils (Type III tank) to a hydrostatic pressure of 1,000 psi for not less than 10 minutes. Evidence of leakage or permanent deformation shall constitute failure of this test.

4.7 Inspection comparison. The Government may select tanks at any time during the contract production period and subject these tanks to the examination specified in 4.6.1 and the tests specified in 4.6.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government on units selected at random from those which have been accepted by the Government and will not include the previously inspected first produced tanks and initial production tanks. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison tank to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison tanks and shall be cause for refusal by the Government to continue acceptance of production tanks until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on tanks previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all tanks accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Preproduction pack inspection.

Source: https://assist.dla.mil -- Downloaded: 202610-17T13:32Z
Check the source to verify that this is the current version before use.

4.8.1.1 Examination. The preproduction pack shall be examined for the defects specified in 4.8.2.3. Presence of one or more defects shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection a completed pack prepared for shipment shall be considered a unit of product.

4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

113.	Materials, methods, or containers not as specified for Level A. Each incorrect material, method, or container shall constitute one defect.
114.	Structural frame members of unlike description boxed or bundled together for Level A.
115.	Fiberboard boxes not closed and sealed by the method specified for Level A.
116.	Bristles of brushes not wrapped in accordance with the refere. document for Level A.
117.	Sealing compound not preserved in accordance with the referenced document for Level A.
118.	Putty not preserved in accordance with the referenced document for Level A.
119.	Asphalt tank heating coils not preserved as specified for Level A.
120.	Consolidation not as specified for Level A.
121.	Packing not as specified, or as required by applicable drawings for Level A.
122.	Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.12). The preproduction

Source: https://assist.dla.mil -- Downloaded: 202610-17T13:32Z
Check the source to verify that this is the currentversion before use.

pack may be accomplished utilizing either the first produced tank or the production tanks. If the first produced tank is utilized, any preservation, and packing shall be removed by the contractor at no expense to the Government when requested by the Government to facilitate comparison between the first produced tank and the production tanks.

5.2 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in, and shall be applied in accordance with MIL-P-116.

5.2.1.2 Structural frame members. Structural frame members of like description shall be either boxed or bundled together in quantities shown on the applicable drawings. Each bundle shall be secured with not less than two evenly spaced flat steel straps conforming to QQ-S-781, Finish B, size as applicable. Strapping used on Army sponsored shipments will conform to QQ-S-781, Finish A.

5.2.1.3 Body staves, bottom, and deck sheets or plates. The body staves, bottom, and deck sheets or plates shall not require preservation.

5.2.1.4 Gasket strips, formed gaskets, and bonded water-gaskets. Gasket strips, formed gaskets, and bonded washer-gaskets of like size and description required for each tank shall be placed in boxes conforming to PPP-B-636, W5c or W6c, as applicable. The boxes shall be closed and sealed as specified for Method V in accordance with the appendix to the box specification.

5.2.1.5 Bolts, with nuts, and replacement plugs. Bolts with nuts, and bolt replacement plugs shall be clean, free from corrosion, and shall have a light oil coating. Bolts with nuts of like description in the quantities shown on the drawings, and bolt replacement plugs in the quantities required for each tank shall be placed in fiberboard boxes conforming to PPP-B-636, V3c. Each box shall be closed and sealed as specified for Method V in accordance with the appendix to the box specification. Prior to placing the bolts, nuts, and plugs in the boxes, they shall be completely enclosed in Type P-18 volatile corrosion inhibitor paper. Cushioning shall be provided to prevent movement within the boxes. The boxes shall be wrapped in barrier material conforming to MIL-B-131, Class 1, with all seams and joints of the barrier material fully heat sealed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

5.2.1.6 Brushes. The bristles of each brush shall be wrapped in accordance with the Level A requirements of H-B-420. The quantity of brushes necessary for each tank shall then be placed together in a close-fitting box conforming to PPP-B-636, W6c, style optional. The box shall be closed and sealed as specified for Method V in accordance with the appendix to the box specification.

5.2.1.7 Sealing compound. The sealing compound shall be preserved in accordance with the Level A requirements of MIL-S-14231.

5.2.1.8 Putty. Putty shall be preserved in accordance with the Level A requirements of MIL-P-20628.

5.2.1.9 Erection manuals. Erection manuals required for each tank shall be preserved in accordance with MIL-P-116, Method IC-1.

5.2.1.10 Fitting sets, automatic liquid-level indicating gage, gage board, and other miscellaneous tank parts. Parts of the fitting sets, automatic liquid-level indicating gage, gage board, and miscellaneous parts not specifically covered herein which require additional protection shall be wrapped with barrier material conforming to PPP-B-1055, Class B-2, secured with tape conforming to PPP-T-60, Type IV, or shall be placed in boxes conforming to PPP-B-636, V3c, and contents cushioned and secured within the box to prevent movement or damage.

5.2.1.11 Asphalt tank heating coils. The heating coils of the asphalt tank shall be preserved in accordance with MIL-P-116, Method I. The interior surfaces of the steam coils shall be coated with Type P-21 preservative and the exterior surfaces of the coils and unprotected components and parts shall be coated with Type P-1 preservative. Open ends of the steam coils shall be protected with threaded metal or plastic caps or plugs or by covering with barrier material conforming to MIL-B-121, Type I, Grade A, Class 2, secured in place with tape conforming to PPP-T-60, Type IV. Parts requiring additional protection shall be preserved as specified in 5.2.1.10.

5.2.1.12 Consolidation. The components of each tank, in the quantities shown on the applicable drawings shall be consolidated, and secured to prevent movement, in snug-fitting boxes conforming to PPP-B-621, Class 1, style optional, or to PPP-B-601, Domestic Type, style optional. Strapping for the boxes shall not be required.

5.2.2 Commercial. Commercial preservation shall be in accordance with MIL-STD-1188.

Source: https://assist.dla.mil -- Downloaded: 202610-17T13:32Z
Check the source to verify that this is the current version before use.

5.3 Packing. Packing shall be Level A or Commercial as specified (see 6.2).

5.3.1 Level A.

5.3.1.1 Steel tanks. The preserved and unpreserved components comprising each complete tank shall be packed as shown on the following drawings for the capacity indicated:

Type I and II

Capacity, barrels	Drawing No.
100	D13217E2317-1 through -6
250	D13217E2319-1 through -5
500	D13217E2320-1 through -7
1,000	D13217E2321-1 through -10
3,000	D13217E2322-1 through -14
10,000	D13217E2323-1 through -20

Type III

100	D13217E2318-1 through -6

5.3.1.2 Arrangement, blocking, bracing, and anchoring. Arrangement within the crates of components comprising each complete tank shall be as shown on the applicable drawings. Blocking and bracing not detailed on drawings shall be in accordance with MIL-STD-1186. Flat steel strapping used for anchoring shall conform to QQ-S-781, Type I, Finish B, size as applicable.

5.3.1.3 Closure and strapping. The crates shall be closed and strapped in accordance with the appendix to the applicable crate specification except straps shall be Finish B.

5.3.1.4 Fitting set. The components comprising each complete fitting set furnished separately from the tank shall be packed in a box conforming to PPP-B-621, Class 2, style optional, or PPP-B-601, Overseas Type, Grade B, style optional. The boxes shall be close-fitting and shall be strapped in accordance with the appendix to the applicable box specification. The packed fitting set shall be shipped separately from but at the same time as the tank.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

5.3.1.4.1 Asphalt tank heating coils. All components of TA13216E8260 shall be packed in a crate conforming to MIL-C-104, Type I, Class 1 or 2, Style A. The contents of the crate shall be blocked, braced, and anchored as specified in 5.3.1.2. The packed heating coils and accessories shall be shipped separately from but at the same time as the tank.

5.3.2 Commercial. Commercial packing shall be in accordance with MIL-STD-1188.

5.4 Marking. For military levels of protection, marking shall be in accordance with MIL-STD-129. In addition, the exterior of each shipping container for Type II and III tanks shall be stenciled "DO NOT USE FOR POTABLE WATER" on all four sides in letters as large as practicable. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. Bolted steel tanks are intended for field erection as temporary or semipermanent on-the-ground or tower-supported storage facilities for gasoline, fuel, water, and potable water. The tower fitting sets are used to convert the 100, 250, 500, 1,000 and 3,000 bbl tanks from on-the-ground storage to tower-supported storage facilities.

6.2 Ordering data. Procurement documents should specify the following.

(a) Title, number and date of this specification.
(b) Type and capacity of tank required (see 1.2).
(c) Time frame required for submission of the first produced tank; and number of tanks to be furnished (see 3.2).
(d) When initial production inspection is required and the number of tanks to be furnished (see 3.3).
(e) When stepped type gaskets (applicable to Navy procurements only) are to be furnished (see 3.5 and 6.4).
(f) When sealing compound, putty, and brushes shall be furnished with the tank (see 3.6 and 6.5).
(g) Technical publications required (see 3.11 and 6.6).
(h) When tower fitting sets are required (see 3.13.1).
(i) When the interior of Type II tanks shall be finished other than as specified in 3.9.2.2.2 and 3.13.3.
(j) When the interior of Type III tanks shall be coated (see 3.14).
(k) When the use of Government designed inspection equipment is not required for the 500 bbl, 1000 bbl, 3000 bbl, and 10,000 bbl capacity tanks (see 3.15 and 6.11).

Source: https://assist.dla.mil -- Downloaded: 202610-17T13:32Z
Check the source to verify that this is the current version before use.

(l) Whether Government designed inspection equipment will be loaned by the Government or shall be fabricated and furnished by the contractor (see 3.15 and 6.11).
(m) Time frame required for submission of the preproduction pack (see 5.1).
(n) Degree of preservation and degree of packing required (see 5.2 and 5.3).

6.3 First produced tank. Any changes or deviations of production tanks from the approved first produced tank during production will be subject to the approval of the contracting officer. Approval of the first produced tank will not relieve the contractor of his obligation to furnish tanks conforming to this specification.

6.4 Stepped type gaskets. Stepped type gaskets should be specified for Navy procurement only.

6.5 Sealing compound and putty. Care must be exercised when ordering sealing compound and putty since both of these items are perishable. For Army use, when the tanks are intended for immediate service, the sealing compound, putty and brushes should be ordered concurrently with the tanks but packaged and shipped separately from the tanks. When the tanks are intended to be placed in depot storage, the sealing compound, putty, and brushes should not be ordered and placed in storage with the tanks. Normally, tanks for Navy use will be procured without brushes but with sealing compound and putty packaged with the tank.

WARNING: Sealing compound and putty contain toxic materials and are not to be ordered when the tanks are to be used for potable water storage.

6.6 Government-furnished property. The contracting officer should arrange to furnish the following manual (see 3.11):

TM-5-5430-209-12 or TO 36Y31-1-101	- Tank, Steel, Vertical, Bolted, Knock-down, Sealed, Openings Standard Bottom and Roof, Gasoline, Oil, or Water.

6.7 Government-loaned property. The contracting officer should arrange to loan, when applicable, the Government inspection equipment specified in 3.15.

Source: https://assist.dla.mil -- Downloaded: 202610-17T13:32Z
Check the source to verify that this is the current version before use.

6.8 Materials for Type I tank. The contracting officer should require the contractor to furnish certification that all materials used which may come in contact with potable water have been approved by the appropriate Department medical service officer (see 3.4); or are listed in the current edition of the List of Accepted Equipment for Interstate Carrier Use (see 3.13.2).

6.9 Quality assurance provisions (QAP). The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable.

6.10 Definition. A QAP is a contractual requirement that supplements section 4 of the specification. QAP's indicate the minimum requirements which must be inspected on the product drawings to verify design objectives of the product and assure interchangeability of repair parts.

6.11 Government inspection equipment design for 500 bbl, 1000 bbl, 3000 bbl, and 10,000 bbl capacity tanks. Government inspection equipment design drawings exist for these tanks and are listed on the applicable TA's. When this inspection equipment is not used to inspect these tanks, the erection test specified in 4.6.2.1 is required.

6.12 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, package, pack, and mark the tanks in accordance with this specification.

6.13 International agreement. Provisions of this specification are the subject of international standardization agreement (NATO, STANAG No. 2761) (ABC ST 45). When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective departmental standardization offices so that appropriate action may be taken respecting the international agreement concerned.

6.14 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

Custodians:
Army - ME
Navy - YD

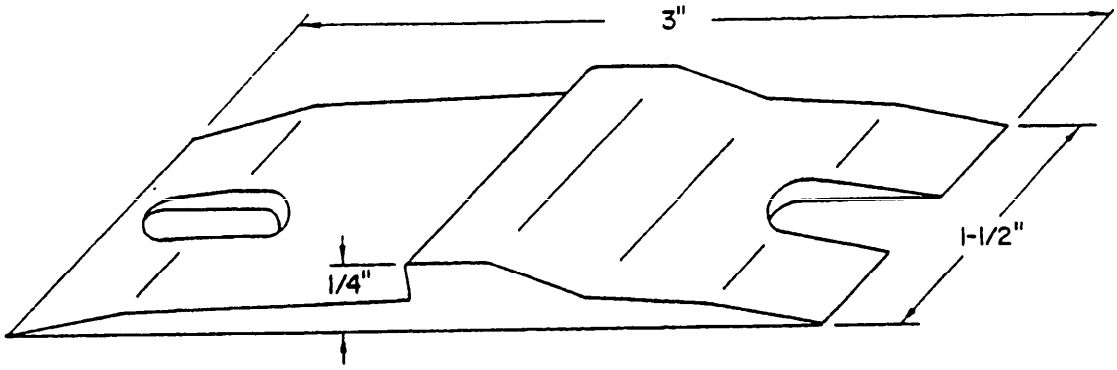
Review activities:
Army - MD
DLA - CS

User activities:
Navy - MC

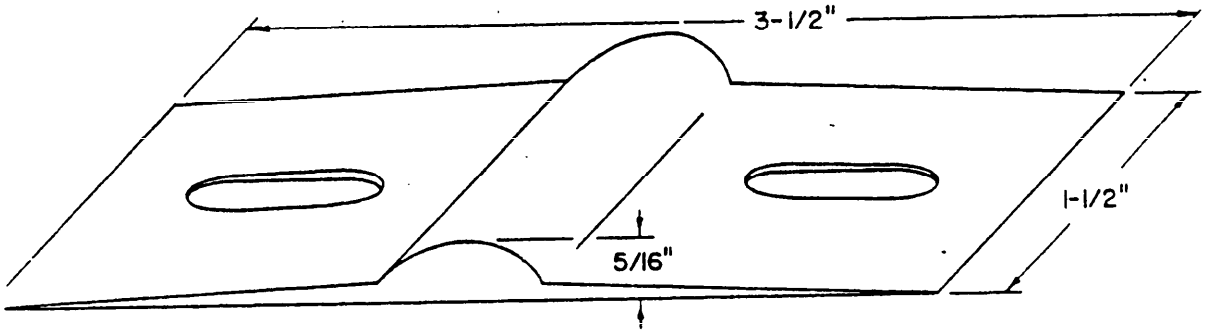
Preparing activity:
Army - ME

Project 5430-0128

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.



STEP TYPE SEALING GASKET



OFFSET TYPE SEALING GASKET

NOTE UNLESS OTHERWISE SPECIFIED, DIMENSIONAL TOLERANCES SHALL BE AS SPECIFIED IN MIL-G-1086 TYPE II

FIGURE 1. Sealing gasket variations.

(FOR NAVY USE ONLY)

X1923A

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-T-10086H Tanks, Liquid Storage, Metal: Vertical, Bolted

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Mobility Equipment Research and
Development Command
DRDME-DE/DS
Fort Belvoir, VA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.